

Cross-Dataset Generalization of Transformer-Based Intrusion Detection Systems for the Internet of Things

A Research Area Brief

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September 15, 2026

1 Context

The broad Computer Science area of this work is *machine learning for cybersecurity*, and more specifically the use of deep learning and Transformer models to build Intrusion Detection Systems (IDS) for the Internet of Things (IoT). An IDS monitors network or device activity and raises alerts when it observes behaviour consistent with an attack. This area matters to a technically informed reader for a concrete reason: the number of connected IoT devices, and therefore the attack surface exposed to adversaries, has grown rapidly, while many IoT devices are too constrained to run traditional, signature-based defences [1]. As a result, data-driven detection based on machine learning, and increasingly on attention-based Transformer architectures, has become a central research direction for protecting modern IoT deployments. Recent work applies Transformer encoders directly to flow-level IoT traffic and reports competitive multi-class detection performance on contemporary datasets [2], which establishes both the technical relevance and the momentum of the area.

2 Problem Area

The problem addressed here is not a missing technology but a gap in what is currently *known* about how well these models actually work once deployed. The dominant evaluation practice in the literature is to train and test a model on a single dataset, where reported accuracy is often very high (frequently above 99%). What remains uncertain, and arguably unsafe, is whether such a model generalizes to a *different* IoT environment—one with different devices, traffic distributions, and attack taxonomies than the dataset it was trained on. Evidence indicates that it often does not: a recent dataset-centric evaluation reports degradation of up to roughly 30 percentage points in macro-F1 when models trained on one modern IoT dataset are tested on unseen ones [1]. This creates a practical risk: a detector that appears excellent in the laboratory may fail silently after deployment, missing attacks in a real network.

Those who experience this problem, or benefit from solving it, include the security teams and operators who deploy IDS in real IoT networks (for example smart homes, industrial IoT, and healthcare IoT), the device and platform vendors who rely on these detectors, and the end users whose safety depends on the protection working outside the exact conditions in which it was measured. Researchers also benefit, because trustworthy cross-environment evaluation is a precondition for credible progress in the field.

3 Possible Contribution

This brief describes two plausible research contributions and the minimum credible evidence each would require.

3.1 Possible Contribution 1: An empirical finding and evaluation protocol

The first contribution is an *empirical finding*: a controlled, reproducible measurement of how much Transformer-based IDS models degrade when they are trained on one modern IoT dataset and evaluated on others (for example CIC-IoT2023, TON_IoT, Edge-IIoTset, and RT_IoT2022), under a harmonized feature and label protocol. The minimum credible evidence would be a documented experimental pipeline; multiple datasets with aligned feature spaces and label taxonomies; clearly specified baseline models; and a report of in-domain versus cross-dataset performance using metrics that are meaningful under class imbalance (macro-F1, recall, and false-positive rate rather than accuracy alone). The finding is credible only if the degradation is observed consistently across reasonable model and configuration choices, not for a single lucky or unlucky setting.

3.2 Possible Contribution 2: A method that improves cross-dataset robustness

The second contribution is an *artefact or training strategy*: a Transformer-based IDS approach—for example multi-dataset training, feature harmonization, or lightweight domain adaptation—that measurably reduces the cross-dataset degradation relative to single-dataset-trained baselines. The minimum credible evidence would be a demonstration that the proposed method retains higher macro-F1 on *unseen* datasets than the baselines under the same protocol; an ablation isolating which component of the method drives the improvement; and a bounded claim scoped to the datasets and attack families actually tested. Crucially, the evidence must show improvement in out-of-domain performance, not merely a higher in-domain accuracy, since higher in-domain accuracy is exactly the misleading signal the first contribution warns against.

4 Research Rather Than Routine Development

This work would create defensible knowledge rather than only software or a client-specific configuration. Following the required logic:

Meaningful problem → answerable question → valuable contribution → appropriate method → credible evidence → bounded claim.

The *meaningful problem* is the silent failure of high-accuracy IDS models on network environments they were not trained on. The *answerable question* is: how much do Transformer-based IDS models degrade under cross-dataset evaluation, and can a defined training or adaptation strategy reduce that degradation? The *valuable contribution* is generalizable evidence about evaluation practice plus a candidate method to improve robustness. The *appropriate method* is a controlled multi-dataset experiment with fixed protocols and baselines. The *credible evidence* is consistent, reproducible metrics reported across datasets and configurations. The *bounded claim* restricts conclusions to the tested datasets and attack families. Because the outcome is a reusable protocol and a tested claim about robustness—verifiable against baselines and reproducible by others—rather than a one-off tool built for a single deployment, it constitutes research rather than routine development.

5 Initial Search Terms

The following search terms and phrases can be refined in later chapters:

- cross-dataset generalization intrusion detection IoT
- transformer intrusion detection IoT
- domain adaptation network intrusion detection

Suitable Digital Libraries

Two digital libraries are well suited to this area and are accessible through the Saudi Digital Library:

- **IEEE Xplore** — strong coverage of networking, security, and IoT conference and journal papers.
- **ScienceDirect (Elsevier)** — broad coverage of computers and security and computer networks journals.

6 Responsible Use of Generative AI

A generative AI assistant was used to support this brief in a limited and disclosed way: to brainstorm and sharpen the framing of the research angle, to translate and summarize the abstracts of English-language sources, and to draft candidate search terms. Every scholarly source cited here was then verified independently against the original publication (title, authors, venue, year, and DOI) before inclusion, and the argument was reviewed and edited by the author. No confidential, personal, or restricted research information was entered into any external AI service.

References

- [1] M. A. Bilal, I. Ul Islam, S. Idrees, M. Qasim, M. J. Khan, and J. Khan, “Dataset-centric evaluation of federated intrusion detection models in IoT networks,” *Scientific Reports*, vol. 16, no. 1, p. 2683, 2026. DOI: [10.1038/s41598-025-32567-w](https://doi.org/10.1038/s41598-025-32567-w)
- [2] S.-M. Tseng, Y.-Q. Wang, and Y.-C. Wang, “Multi-class intrusion detection based on transformer for IoT networks using the CIC-IoT-2023 dataset,” *Future Internet*, vol. 16, no. 8, p. 284, 2024. DOI: [10.3390/fi16080284](https://doi.org/10.3390/fi16080284)